financial_services

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correlation_matrix

univariate_analysis

customer_residence_citizenship

Distribution of Customer State of Residence

80 — 2.5K

Frequency

Distribution of Customer Citizenship

160 — 1.9K

Frequency

customer_demographics

Distribution of Gender, Marital Status

What are the characteristics of Response?

event_indicator ranges from 0 to 1. Average event_indicator is 0.3. Most cases (the middle 80%) have an event_indicator between 0 and 1. net_worth best differentiates the highest (top 10%) and the lowest (bottom 10%) event_indicator cases. The three most related factors are net_worth, credit_score, and num_dependents.

What factors are most related to event_indicator?

What is the relationship between Measure and Category?

When at_current_job_1_year is 0, the average of event_indicator is 0.54.
When at_current_job_1_year is 1, the average of event_indicator is 0.05.
The most common at_current_job_1_year value is 0.

decision_tree

Decision Tree **event_indicator** Fit: **ASE 0.04** Observations: **15K of 15K**

Tree

Variable Importance

Assessment

event_indicator

logistic_regression

Logistic Regression **event_indicator** Event: **1** Fit: **Validation KS (Youden) 0.8613** Observations: **15K of 15K**

Fit Summary

Residual Plot

Confusion Matrix

cluster_analysis

Cluster Observations: **15K of 15K** Polylines: **8**

What values for the most important factors should be used to predict?

net_worth
-0.375310595

credit_score
0.1863434945

num_dependents
0

at_current_job_1_year
0

credit_history_mos
0.316453002

job_in_education
0

num_transactions
0.0414180523

debt_to_income
-0.011196477

amount
-0.507832251

gender
0

What is the prediction for Response?

0.5446094384

The predicted event_indicator for this case is 81.51% higher than the observed average event_indicator of 0.3. Most observations (70%) have a lower event_indicator than this predicted case. The prediction is based on an automatically selected Gradient Boosting model.

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Histogram - amount 1 Results

amount (lower)	amount (upper)	Frequency
-0.85	-0.75	2,006
-0.75	-0.65	2,777
-0.65	-0.55	2,028
-0.55	-0.45	1,732
-0.45	-0.35	521
-0.35	-0.25	515
-0.25	-0.15	553
-0.15	-0.05	344
-0.05	0.05	208
0.05	0.15	231
0.15	0.25	238
0.25	0.35	242
0.35	0.45	218
0.45	0.55	197
0.55	0.65	238
0.65	0.75	201
0.75	0.85	222
0.85	0.95	218
0.95	1.05	136
1.05	1.15	134
1.15	1.25	119
1.25	1.35	146
1.35	1.45	123
1.45	1.55	110
1.55	1.65	124
1.65	1.75	134
1.75	1.85	130
1.85	1.95	121
1.95	2.05	124
2.05	2.15	169
2.15	2.25	152
2.25	2.35	169
2.35	2.45	177
2.45	2.55	149
2.55	2.65	191
2.65	2.75	167
2.75	2.85	87

Histogram - credit_score 1 Results

credit_score (lower)	credit_score (upper)	Frequency
-5	-4.8	2
-4.8	-4.6	4
-4.6	-4.4	10
-4.4	-4.2	5
-4.2	-4	2
-4	-3.8	4
-3.8	-3.6	31
-3.6	-3.4	50
-3.4	-3.2	77
-3.2	-3	60
-3	-2.8	50
-2.8	-2.6	53
-2.6	-2.4	158
-2.4	-2.2	167
-2.2	-2	201
-2	-1.8	174
-1.8	-1.6	188
-1.6	-1.4	186
-1.4	-1.2	359
-1.2	-1	418
-1	-0.8	445
-0.8	-0.6	664
-0.6	-0.4	754
-0.4	-0.2	700
-0.2	0	1,469
0	0.2	1,540
0.2	0.4	1,511
0.4	0.6	1,545
0.6	0.8	1,483
0.8	1	1,444
1	1.2	558
1.2	1.4	561
1.4	1.6	478

Histogram - net_worth 1 Results

net_worth (lower)	net_worth (upper)	Frequency
-1.05	-0.9	746
-0.9	-0.75	1,818
-0.75	-0.6	1,961
-0.6	-0.45	2,093
-0.45	-0.3	1,571
-0.3	-0.15	788
-0.15	0	754
0	0.15	846
0.15	0.3	769
0.3	0.45	820
0.45	0.6	556
0.6	0.75	263
0.75	0.9	274
0.9	1.05	282
1.05	1.2	261
1.2	1.35	213
1.35	1.5	225
1.5	1.65	209
1.65	1.8	141
1.8	1.95	39
1.95	2.1	38
2.1	2.25	48
2.25	2.4	31
2.4	2.55	39
2.55	2.7	31
2.7	2.85	46
2.85	3	37
3	3.15	30
3.15	3.3	47
3.3	3.45	31
3.45	3.6	33
3.6	3.75	41
3.75	3.9	38
3.9	4.05	35
4.05	4.2	46
4.2	4.35	56
4.35	4.5	46
4.5	4.65	32
4.65	4.8	17

Histogram - num_dependents 1 Results

num_dependents (lower)	num_dependents (upper)	Frequency
0	1	4,079
1	2	2,834
2	3	2,417
3	4	1,726
4	5	1,581
5	6	1,071
6	7	1,121
7	8	522

Histogram - age 1 Results

age (lower)	age (upper)	Frequency
-1.56	-1.44	625
-1.44	-1.32	569
-1.32	-1.2	591
-1.2	-1.08	312
-1.08	-0.96	624
-0.96	-0.84	661
-0.84	-0.72	661
-0.72	-0.6	639
-0.6	-0.48	653
-0.48	-0.36	487
-0.36	-0.24	1,038
-0.24	-0.12	1,006
-0.12	0	991
0	0.12	1,057
0.12	0.24	671
0.24	0.36	185
0.36	0.48	357
0.48	0.6	329
0.6	0.72	380
0.72	0.84	332
0.84	0.96	289
0.96	1.08	146
1.08	1.2	291
1.2	1.32	288
1.32	1.44	330
1.44	1.56	301
1.56	1.68	282
1.68	1.8	150
1.8	1.92	265
1.92	2.04	212
2.04	2.16	209
2.16	2.28	157
2.28	2.4	171
2.4	2.52	92

Histogram - num_transactions 1 Results

num_transactions (lower)	num_transactions (upper)	Frequency
-1.92	-1.8	286
-1.8	-1.68	284
-1.68	-1.56	258
-1.56	-1.44	665
-1.44	-1.32	485
-1.32	-1.2	347
-1.2	-1.08	369
-1.08	-0.96	526
-0.96	-0.84	354
-0.84	-0.72	372
-0.72	-0.6	407
-0.6	-0.48	632
-0.48	-0.36	421
-0.36	-0.24	451
-0.24	-0.12	695
-0.12	0	1,004
0	0.12	701
0.12	0.24	693
0.24	0.36	708
0.36	0.48	950
0.48	0.6	603
0.6	0.72	594
0.72	0.84	772
0.84	0.96	223
0.96	1.08	238
1.08	1.2	229
1.2	1.32	361
1.32	1.44	207
1.44	1.56	225
1.56	1.68	253
1.68	1.8	370
1.8	1.92	220
1.92	2.04	209
2.04	2.16	239

Histogram - debt_to_income 1 Results

debt_to_income (lower)	debt_to_income (upper)	Frequency
-1.86	-1.74	349
-1.74	-1.62	416
-1.62	-1.5	380
-1.5	-1.38	416
-1.38	-1.26	509
-1.26	-1.14	450
-1.14	-1.02	421
-1.02	-0.9	423
-0.9	-0.78	526
-0.78	-0.66	526
-0.66	-0.54	510
-0.54	-0.42	527
-0.42	-0.3	707
-0.3	-0.18	532
-0.18	-0.06	622
-0.06	0.06	633
0.06	0.18	851
0.18	0.3	647
0.3	0.42	561
0.42	0.54	429
0.54	0.66	607
0.66	0.78	439
0.78	0.9	416
0.9	1.02	433
1.02	1.14	557
1.14	1.26	447
1.26	1.38	473
1.38	1.5	522
1.5	1.62	286
1.62	1.74	258
1.74	1.86	273
1.86	1.98	54
1.98	2.1	37
2.1	2.22	43
2.22	2.34	40
2.34	2.46	31

Histogram - credit_history_mos 1 Results

credit_history_mos (lower)	credit_history_mos (upper)	Frequency
-3.42	-3.3	116
-3.3	-3.18	117
-3.18	-3.06	90
-3.06	-2.94	38
-2.94	-2.82	67
-2.82	-2.7	35
-2.7	-2.58	58
-2.58	-2.46	35
-2.46	-2.34	104
-2.34	-2.22	78
-2.22	-2.1	122
-2.1	-1.98	74
-1.98	-1.86	112
-1.86	-1.74	85
-1.74	-1.62	117
-1.62	-1.5	80
-1.5	-1.38	110
-1.38	-1.26	66
-1.26	-1.14	183
-1.14	-1.02	430
-1.02	-0.9	288
-0.9	-0.78	423
-0.78	-0.66	331
-0.66	-0.54	611
-0.54	-0.42	392
-0.42	-0.3	570
-0.3	-0.18	428
-0.18	-0.06	610
-0.06	0.06	411
0.06	0.18	670
0.18	0.3	635
0.3	0.42	1,369
0.42	0.54	936
0.54	0.66	1,380
0.66	0.78	804
0.78	0.9	1,137
0.9	1.02	761
1.02	1.14	747
1.14	1.26	731

Geo region - region_char 1 Region

region_char	Frequency
Idaho	80
Colorado	86
Wyoming	91
Oregon	91
Wisconsin	94
Michigan	95
Missouri	98
Montana	104
Nevada	105
Indiana	108
Ohio	108
Minnesota	109
Arizona	111
Washington	113
Illinois	116
Kansas	117
Nebraska	117
Utah	119
Iowa	203
Connecticut	223
Rhode Island	226
Tennessee	233
New Jersey	237
Delaware	238
Alabama	242
Kentucky	247
Georgia	247
Mississippi	248
Arkansas	250
Maryland	255
Virginia	259
Maine	259
South Carolina	260
North Carolina	262
Vermont	267
West Virginia	275
New Hampshire	280
Massachusetts	280
Pennsylvania	287
New York	1,835
California	1,836
Florida	2,055
Texas	2,485

Geo region - citizenship_char 1 Region

citizenship_char	Frequency
Bolivia	160
Peru	164
Costa Rica	167
United Kingdom	171
Sweden	176
Uruguay	178
Brazil	182
Argentina	182
France	184
Norway	185
Panama	187
Paraguay	193
Denmark	195
Spain	196
Chile	199
Germany	203
Colombia	204
Ireland	206
Portugal	208
Italy	208
Ethiopia	223
Botswana	225
Tunisia	231
Tanzania	231
Morocco	238
Zimbabwe	240
South Africa	241
Egypt	242
Angola	243
Sudan	261
Canada	1,800
China	1,821
United States	1,850
Mexico	1,871
India	1,886

Pie - gender_char 1 Results

gender_char	Frequency	marital_status_char
female	1,090	Divorced
male	2,347	Divorced
female	3,107	Married
male	6,906	Married
female	512	Single
male	1,389	Single

Automated explanation - event_indicator 1 Explanation Description

1. Select response for Automated Explanation	A report author selected event_indicator as the response.
2. Screen factors	Automated Explanation modified or removed 35 of 43 factors. See the Screening Results tab for details.
3. Determine most related factors	Automated Explanation used a one-level decision tree for each factor to determine its relative importance to event_indicator. For example, the input credit_score has a relative importance of 0.76 which means it is 0.76 times as important as net_worth.

Automated explanation - event_indicator 1 Screening Results

Factor	Action Taken
age	
amount	
at_current_job_1_year	The measure was converted to category because relatively few distinct levels were detected.
business_owner	The measure was converted to category because relatively few distinct levels were detected.
citizenship_africa	The measure was converted to category because relatively few distinct levels were detected.
citizenship_china	The measure was converted to category because relatively few distinct levels were detected.
citizenship_eu	The measure was converted to category because relatively few distinct levels were detected.
citizenship_india	The measure was converted to category because relatively few distinct levels were detected.
citizenship_latam	The measure was converted to category because relatively few distinct levels were detected.
citizenship_middle_east	The measure was converted to category because relatively few distinct levels were detected.
citizenship_other	The measure was converted to category because relatively few distinct levels were detected.
citizenship_us	The measure was converted to category because relatively few distinct levels were detected.
credit_history_mos	
credit_score	
debt_to_income	
ever_missed_obligation	The measure was converted to category because relatively few distinct levels were detected.
gender	The measure was converted to category because relatively few distinct levels were detected.
homeowner	The measure was converted to category because relatively few distinct levels were detected.
job_in_education	The measure was converted to category because relatively few distinct levels were detected.
job_in_energy	The measure was converted to category because relatively few distinct levels were detected.
job_in_financial	The measure was converted to category because relatively few distinct levels were detected.
job_in_healthcare	The measure was converted to category because relatively few distinct levels were detected.
job_in_hospitality	The measure was converted to category because relatively few distinct levels were detected.
job_in_manufacturing	The measure was converted to category because relatively few distinct levels were detected.
job_in_media	The measure was converted to category because relatively few distinct levels were detected.
job_in_transport	The measure was converted to category because relatively few distinct levels were detected.
marital_status_divorced	The measure was converted to category because relatively few distinct levels were detected.
marital_status_married	The measure was converted to category because relatively few distinct levels were detected.
marital_status_single	The measure was converted to category because relatively few distinct levels were detected.
net_worth	
num_dependents	The measure was converted to category because relatively few distinct levels were detected.
num_transactions	
region_ca	The measure was converted to category because relatively few distinct levels were detected.
region_fl	The measure was converted to category because relatively few distinct levels were detected.
region_mw	The measure was converted to category because relatively few distinct levels were detected.
region_ne	The measure was converted to category because relatively few distinct levels were detected.
region_ny	The measure was converted to category because relatively few distinct levels were detected.
region_so	The measure was converted to category because relatively few distinct levels were detected.
region_tx	The measure was converted to category because relatively few distinct levels were detected.
region_we	The measure was converted to category because relatively few distinct levels were detected.
smoker	The measure was converted to category because relatively few distinct levels were detected.
uses_direct_deposit	The measure was converted to category because relatively few distinct levels were detected.
years_at_residence	

Automated explanation - event_indicator 1 Relative Importance

Factor	Importance
net_worth	1.0000
credit_score	0.7616
num_dependents	0.7261
at_current_job_1_year	0.7113
credit_history_mos	0.4723
job_in_education	0.4575
num_transactions	0.3014
debt_to_income	0.2681
amount	0.2391
gender	0.1768
age	0.1193
job_in_hospitality	0.1039
years_at_residence	0.0882
homeowner	0.0862
job_in_financial	0.0647
uses_direct_deposit	0.0584
job_in_manufacturing	0.0560
smoker	0.0500
region_ca	0.0462
region_ny	0.0423
job_in_healthcare	0.0386
marital_status_single	0.0367
business_owner	0.0365
ever_missed_obligation	0.0254
region_mw	0.0213
job_in_transport	0.0208
marital_status_married	0.0204
region_we	0.0174
job_in_media	0.0132
region_tx	0.0104
job_in_energy	0.0100
region_fl	0.0096
region_ne	0.0082
region_so	0.0064
citizenship_eu	0.0003
citizenship_other	0.0002
citizenship_middle_east	0.0002
marital_status_divorced	0.0001
citizenship_us	0.0001
citizenship_china	0.0001
citizenship_africa	0.0001
citizenship_latam	0.0000
citizenship_india	0.0000

Automated explanation - event_indicator 1 Anomalies

event_indicator	Location
No anomalies detected	

Decision tree - event_indicator 1 Node Statistics

Node ID	Depth	Parent ID	N Children	Type	Observations	% Observations	Percent of Parent	N Missing	Gain	Predicted Value	Average	Std. Dev
0	0	-1	2	Class	15,351	100.00%	.	0	1,275.8747	0.3000455996	0.3000455996	0.4583
1	1	0	2	Class	7,982	52.00%	52.00%	0	726.9317	0.5770483588	0.5770483588	0.4941
2	1	0	0	Leaf	7,369	48.00%	48.00%	0	0.0000	0	0	0.0000
3	2	1	2	Class	5,411	35.25%	67.79%	0	334.9431	0.7850674552	0.7850674552	0.4108
4	2	1	2	Class	2,571	16.75%	32.21%	0	53.9593	0.1392454298	0.1392454298	0.3463
5	3	3	2	Class	4,647	30.27%	85.88%	0	95.8179	0.8859479234	0.8859479234	0.3179
6	3	3	2	Class	764	4.98%	14.12%	0	33.8035	0.1714659686	0.1714659686	0.3772
7	3	4	2	Class	557	3.63%	21.66%	0	43.5025	0.4147217235	0.4147217235	0.4931
8	3	4	2	Class	2,014	13.12%	78.34%	0	12.4429	0.0630585899	0.0630585899	0.2431
9	4	5	2	Class	4,055	26.42%	87.26%	0	15.9675	0.9408138101	0.9408138101	0.2360
10	4	5	2	Class	592	3.86%	12.74%	0	31.6803	0.5101351351	0.5101351351	0.5003
11	4	6	2	Class	305	1.99%	39.92%	0	30.3507	0.4295081967	0.4295081967	0.4958
12	4	6	0	Leaf	459	2.99%	60.08%	0	0.0000	0	0	0.0000
13	4	7	2	Class	367	2.39%	65.89%	0	16.5962	0.6158038147	0.6158038147	0.4871
14	4	7	2	Class	190	1.24%	34.11%	0	0.4899	0.0263157895	0.0263157895	0.1605
15	4	8	2	Class	327	2.13%	16.24%	0	10.0482	0.2415902141	0.2415902141	0.4287
16	4	8	2	Class	1,687	10.99%	83.76%	0	2.3914	0.0284528749	0.0284528749	0.1663
17	5	9	0	Leaf	3,064	19.96%	75.56%	0	0.0000	0.9765013055	0.9765013055	0.1515
18	5	9	0	Leaf	991	6.46%	24.44%	0	0.0000	0.8304742684	0.8304742684	0.3754
19	5	10	0	Leaf	319	2.08%	53.89%	0	0.0000	0.724137931	0.724137931	0.4476
20	5	10	0	Leaf	273	1.78%	46.11%	0	0.0000	0.2600732601	0.2600732601	0.4395
21	5	11	0	Leaf	181	1.18%	59.34%	0	0.0000	0.6906077348	0.6906077348	0.4635
22	5	11	0	Leaf	124	0.81%	40.66%	0	0.0000	0.0483870968	0.0483870968	0.2155
23	5	13	0	Leaf	190	1.24%	51.77%	0	0.0000	0.8210526316	0.8210526316	0.3843
24	5	13	0	Leaf	177	1.15%	48.23%	0	0.0000	0.395480226	0.395480226	0.4903
25	5	14	0	Leaf	164	1.07%	86.32%	0	0.0000	0.006097561	0.006097561	0.0781
26	5	14	0	Leaf	26	0.17%	13.68%	0	0.0000	0.1538461538	0.1538461538	0.3679
27	5	15	0	Leaf	288	1.88%	88.07%	0	0.0000	0.1770833333	0.1770833333	0.3824
28	5	15	0	Leaf	39	0.25%	11.93%	0	0.0000	0.7179487179	0.7179487179	0.4559
29	5	16	0	Leaf	1,575	10.26%	93.36%	0	0.0000	0.0184126984	0.0184126984	0.1345
30	5	16	0	Leaf	112	0.73%	6.64%	0	0.0000	0.1696428571	0.1696428571	0.3770

Decision tree - event_indicator 1 Node Rules

Node ID	Parent ID	Type	age	amount	at_current_job_1_year	business_owner	citizenship_africa	citizenship_china	citizenship_eu	citizenship_india	citizenship_latam	citizenship_middle_east	citizenship_other	citizenship_us	credit_history_mos	credit_score	debt_to_income	ever_missed_obligation	gender	homeowner	job_in_education	job_in_energy	job_in_financial	job_in_healthcare	job_in_hospitality	job_in_manufacturing	job_in_media	job_in_transport	marital_status_divorced	marital_status_married	marital_status_single	net_worth	num_dependents	num_transactions	region_ca	region_fl	region_mw	region_ne	region_ny	region_so	region_tx	region_we	smoker	uses_direct_deposit	years_at_residence
0	-1	Class																																												
1	0	Class																														< -0.34 or Missing														
2	0	Leaf																														≥ -0.34														
3	1	Class			< 1 or Missing																											< -0.34 or Missing														
4	1	Class			≥ 1																											< -0.34 or Missing														
5	3	Class			< 1 or Missing																											< -0.34 or Missing	< 3 or Missing													
6	3	Class			< 1 or Missing																											< -0.34 or Missing	≥ 3													
7	4	Class			≥ 1											< -0.236																< -0.34 or Missing														
8	4	Class			≥ 1											≥ -0.236 or Missing																< -0.34 or Missing														
9	5	Class			< 1 or Missing											< 0.397 or Missing																< -0.34 or Missing	< 3 or Missing													
10	5	Class			< 1 or Missing											≥ 0.397																< -0.34 or Missing	< 3 or Missing													
11	6	Class			< 1 or Missing																											< -0.34 or Missing	3 to 4													
12	6	Leaf			< 1 or Missing																											< -0.34 or Missing	≥ 4													
13	7	Class			≥ 1											< -0.236																< -0.34 or Missing	< 3 or Missing													
14	7	Class			≥ 1											< -0.236																< -0.34 or Missing	≥ 3													
15	8	Class			≥ 1											≥ -0.236 or Missing																< -0.34 or Missing	< 1													
16	8	Class			≥ 1											≥ -0.236 or Missing																< -0.34 or Missing	≥ 1 or Missing													
17	9	Leaf			< 1 or Missing										< 0.268 or Missing	< 0.397 or Missing																< -0.34 or Missing	< 3 or Missing													
18	9	Leaf			< 1 or Missing										≥ 0.268	< 0.397 or Missing																< -0.34 or Missing	< 3 or Missing													
19	10	Leaf			< 1 or Missing										< 0.268 or Missing	≥ 0.397																< -0.34 or Missing	< 3 or Missing													
20	10	Leaf			< 1 or Missing										≥ 0.268	≥ 0.397																< -0.34 or Missing	< 3 or Missing													
21	11	Leaf			< 1 or Missing											< 0.268 or Missing																< -0.34 or Missing	3 to 4													
22	11	Leaf			< 1 or Missing											≥ 0.268																< -0.34 or Missing	3 to 4													
23	13	Leaf			≥ 1										< 0.268 or Missing	< -0.236																< -0.34 or Missing	< 3 or Missing													
24	13	Leaf			≥ 1										≥ 0.268	< -0.236																< -0.34 or Missing	< 3 or Missing													
25	14	Leaf			≥ 1											< -0.236															< 1 or Missing	< -0.34 or Missing	≥ 3													
26	14	Leaf			≥ 1											< -0.236															≥ 1	< -0.34 or Missing	≥ 3													
27	15	Leaf			≥ 1											≥ -0.236 or Missing					< 1 or Missing											< -0.34 or Missing	< 1													
28	15	Leaf			≥ 1											≥ -0.236 or Missing					≥ 1											< -0.34 or Missing	< 1													
29	16	Leaf			≥ 1											≥ -0.236 or Missing					< 1 or Missing											< -0.34 or Missing	≥ 1 or Missing													
30	16	Leaf			≥ 1											≥ -0.236 or Missing					≥ 1											< -0.34 or Missing	≥ 1 or Missing													

Decision tree - event_indicator 1 Variable Importance

Variable	Importance
net_worth	1,275.8747
at_current_job_1_year	726.9317
num_dependents	424.6921
credit_score	180.1279
credit_history_mos	64.2440
job_in_education	12.4396
marital_status_single	0.4899
age	0.0000
amount	0.0000
business_owner	0.0000
citizenship_africa	0.0000
citizenship_china	0.0000
citizenship_eu	0.0000
citizenship_india	0.0000
citizenship_latam	0.0000
citizenship_middle_east	0.0000
citizenship_other	0.0000
citizenship_us	0.0000
debt_to_income	0.0000
ever_missed_obligation	0.0000
gender	0.0000
homeowner	0.0000
job_in_energy	0.0000
job_in_financial	0.0000
job_in_healthcare	0.0000
job_in_hospitality	0.0000
job_in_manufacturing	0.0000
job_in_media	0.0000
job_in_transport	0.0000
marital_status_divorced	0.0000
marital_status_married	0.0000
num_transactions	0.0000
region_ca	0.0000
region_fl	0.0000
region_mw	0.0000
region_ne	0.0000
region_ny	0.0000
region_so	0.0000
region_tx	0.0000
region_we	0.0000
smoker	0.0000
uses_direct_deposit	0.0000
years_at_residence	0.0000

Decision tree - event_indicator 1 Assessment

Percentile	Observations	Predicted Average	Observed Average
5	768	0.9765013055	0.9765013055
10	768	0.9765013055	0.9765013055
15	768	0.9765013055	0.9765013055
20	768	0.9749801905	0.9749801905
25	768	0.8304742684	0.8304742684
30	768	0.7773502687	0.7773502687
35	768	0.374599993	0.374599993
40	768	0.0800556486	0.0800556486
45	768	0.0184126984	0.0184126984
50	768	0.0120188492	0.0120188492
55	768	0	0
60	768	0	0
65	768	0	0
70	768	0	0
75	768	0	0
80	768	0	0
85	768	0	0
90	768	0	0
95	768	0	0
100	759	0	0

Decision tree - event_indicator 1 Assessment Statistics

ASE	Observed Average	SSE	Observations Used	Unused
0.0351	0.9765013055	539.1901	15,351	0

Decision tree - event_indicator 1 Assessment Summary

Plot	Summary
Assessment	The data is sorted in descending order by the predicted response for event_indicator. The data is divided into 20 quantiles (with 5% of the data in each), and each quantile reports the average predicted response and average observed response for comparison. For this model, the average predicted and observed responses are equal.

Logistic regression - event_indicator 1 Model Information

Description	Value
Data Source	CASUSER.FINANCIAL_SERVICES_PREP
Response Variable	event_indicator
Distribution	Binary
Link Function	Logit
Optimization Technique	Newton-Raphson with Ridging

Logistic regression - event_indicator 1 Dimensions

Description	Value
Number of Model Effects	6
Number of Classification Effects	0
Number of Columns in X	6
Rank of Cross-product Matrix	6
Observations Read	15,351
Observations Used	15,351
Observations Used for Training	4,605
Observations Used for Validation	1,535

Logistic regression - event_indicator 1 Response Profile

Ordered Value	Count	event_indicator
1	4606	1
2	10745	0

Logistic regression - event_indicator 1 Iteration History

Iteration	Evaluations	Objective	Change	Max Gradient
0	4	0.252757	.	0.173579
1	2	0.192107	0.060649	0.068362
2	2	0.162434	0.029673	0.026078
3	2	0.150826	0.011608	0.008541
4	2	0.148294	0.002533	0.001769
5	2	0.148137	0.000156	0.000115
6	2	0.148137	7.436E-7	5.618E-7
7	2	0.148137	1.87E-11	1.39E-11

Logistic regression - event_indicator 1 Convergence

Reason	Status
Convergence criterion (GCONV=1E-6) satisfied.	0

Logistic regression - event_indicator 1 Fit Statistics

Statistic	Value
-2 Log Likelihood	1364.339
AIC	1376.339
AICC	1376.358
BIC	1414.949
Max-rescaled R-Square	0.856829
R-Square	0.605454

Logistic regression - event_indicator 1 Parameter Estimates

Parameter	Estimate	Standard Error	Chi-Square	Pr > ChiSq
Intercept	-0.39809	0.168606	5.574651	0.01822
net_worth	-4.6268	0.281168	270.7879	<0.00001
credit_score	-1.29114	0.085672	227.128	<0.00001
num_dependents	-1.11539	0.062971	313.7444	<0.00001
at_current_job_1_year	-3.37464	0.175278	370.6825	<0.00001
credit_history_mos	-0.78506	0.077347	103.0182	<0.00001

Logistic regression - event_indicator 1 Type III Test

Effect	DF	Chi-Square	Pr > ChiSq
net_worth	1	270.7879	<0.00001
credit_score	1	227.128	<0.00001
num_dependents	1	313.7444	<0.00001
at_current_job_1_year	1	370.6825	<0.00001
credit_history_mos	1	103.0182	<0.00001

Logistic regression - event_indicator 1 Odds Ratio

Description	Odds Ratio Estimate	Lower Confidence Level (95%) Value	Upper Confidence Level (95%) Value	p-value
at_current_job_1_year	0.03423	0.024278	0.048262	<0.00001
credit_history_mos	0.456094	0.391937	0.530752	<0.00001
credit_score	0.274957	0.232456	0.325228	<0.00001
net_worth	0.009786	0.00564	0.01698	<0.00001
num_dependents	0.327789	0.28973	0.370847	<0.00001

Logistic regression - event_indicator 1 Confusion Matrix

Predicted	Observed	Training Observations	Training Percentage	Validation Observations	Validation Percentage	Test Observations	Test Percentage
0	0	3,067	95.52%	1,008	93.77%	6,099	94.43%
1	0	144	4.48%	67	6.23%	360	5.57%
0	1	119	8.54%	39	8.48%	272	9.88%
1	1	1,275	91.46%	421	91.52%	2,480	90.12%

Logistic regression - event_indicator 1 Lift

Percentile	Training Observations	Training Events	Training Lift	Training Lift Best	Training Cumulative Lift	Training Cumulative Lift Best	Validation Observations	Validation Events	Validation Lift	Validation Lift Best	Validation Cumulative Lift	Validation Cumulative Lift Best	Test Observations	Test Events	Test Lift	Test Lift Best	Test Cumulative Lift	Test Cumulative Lift Best
5.00	231	230	3.2999	3.3142	3.2999	3.3142	77	73	3.1739	3.3478	3.1739	3.3478	461	458	3.3285	3.3503	3.3285	3.3503
10.00	231	225	3.2281	3.3142	3.2640	3.3142	77	76	3.3043	3.3478	3.2391	3.3478	461	451	3.2776	3.3503	3.3031	3.3503
15.00	231	220	3.1564	3.3142	3.2281	3.3142	77	68	2.9565	3.3478	3.1449	3.3478	461	432	3.1395	3.3503	3.2485	3.3503
20.00	231	212	3.0416	3.3142	3.1815	3.3142	77	71	3.0870	3.3478	3.1304	3.3478	461	410	2.9797	3.3503	3.1813	3.3503
25.00	231	202	2.8981	3.3142	3.1248	3.3142	77	62	2.6957	3.3478	3.0435	3.3478	461	384	2.7907	3.3503	3.1032	3.3503
30.00	231	172	2.4677	3.3142	3.0153	3.3142	77	53	2.3043	3.2609	2.9203	3.3333	461	304	2.2093	3.2485	2.9542	3.3333
35.00	231	84	1.2052	0.1148	2.7567	2.8571	77	35	1.5217	0.0000	2.7205	2.8571	461	207	1.5044	0.0000	2.7471	2.8571
40.00	231	32	0.4591	0.0000	2.4695	2.5000	77	14	0.6087	0.0000	2.4565	2.5000	461	72	0.5233	0.0000	2.4691	2.5000
45.00	231	12	0.1722	0.0000	2.2143	2.2222	77	4	0.1739	0.0000	2.2029	2.2222	461	27	0.1962	0.0000	2.2166	2.2222
50.00	231	5	0.0717	0.0000	2.0000	2.0000	77	4	0.1739	0.0000	2.0000	2.0000	461	4	0.0291	0.0000	1.9978	2.0000
55.00	231	0	0.0000	0.0000	1.8182	1.8182	77	0	0.0000	0.0000	1.8182	1.8182	461	2	0.0145	0.0000	1.8175	1.8182
60.00	231	0	0.0000	0.0000	1.6667	1.6667	77	0	0.0000	0.0000	1.6667	1.6667	461	1	0.0073	0.0000	1.6667	1.6667
65.00	231	0	0.0000	0.0000	1.5385	1.5385	77	0	0.0000	0.0000	1.5385	1.5385	461	0	0.0000	0.0000	1.5385	1.5385
70.00	231	0	0.0000	0.0000	1.4286	1.4286	77	0	0.0000	0.0000	1.4286	1.4286	461	0	0.0000	0.0000	1.4286	1.4286
75.00	231	0	0.0000	0.0000	1.3333	1.3333	77	0	0.0000	0.0000	1.3333	1.3333	461	0	0.0000	0.0000	1.3333	1.3333
80.00	231	0	0.0000	0.0000	1.2500	1.2500	77	0	0.0000	0.0000	1.2500	1.2500	461	0	0.0000	0.0000	1.2500	1.2500
85.00	231	0	0.0000	0.0000	1.1765	1.1765	77	0	0.0000	0.0000	1.1765	1.1765	461	0	0.0000	0.0000	1.1765	1.1765
90.00	231	0	0.0000	0.0000	1.1111	1.1111	77	0	0.0000	0.0000	1.1111	1.1111	461	0	0.0000	0.0000	1.1111	1.1111
95.00	231	0	0.0000	0.0000	1.0526	1.0526	77	0	0.0000	0.0000	1.0526	1.0526	461	0	0.0000	0.0000	1.0526	1.0526
100.00	216	0	0.0000	0.0000	1.0000	1.0000	72	0	0.0000	0.0000	1.0000	1.0000	452	0	0.0000	0.0000	1.0000	1.0000

Logistic regression - event_indicator 1 ROC

Cutoff	Training Sensitivity	Training 1 - Specificity	Training KS (Youden)	Validation Sensitivity	Validation 1 - Specificity	Validation KS (Youden)	Test Sensitivity	Test 1 - Specificity	Test KS (Youden)
0.00	1.0000	1.000		1.0000	1.000		1.0000	1.000	
0.01	1.0000	0.295		1.0000	0.308		0.9993	0.310	
0.02	0.9971	0.242		0.9957	0.260		0.9989	0.254	
0.03	0.9964	0.216		0.9935	0.238		0.9982	0.225	
0.04	0.9964	0.198		0.9913	0.217		0.9971	0.204	
0.05	0.9950	0.181		0.9891	0.199		0.9942	0.192	
0.06	0.9943	0.168		0.9891	0.191		0.9927	0.178	
0.07	0.9921	0.157		0.9870	0.180		0.9913	0.167	
0.08	0.9885	0.149		0.9848	0.173		0.9898	0.158	
0.09	0.9871	0.142		0.9848	0.165		0.9880	0.151	
0.10	0.9842	0.137		0.9826	0.154		0.9858	0.143	
0.11	0.9806	0.128		0.9826	0.149		0.9844	0.138	
0.12	0.9806	0.124		0.9826	0.144		0.9826	0.134	
0.13	0.9785	0.119		0.9826	0.137		0.9804	0.128	
0.14	0.9770	0.115		0.9826	0.134		0.9793	0.123	
0.15	0.9756	0.111		0.9804	0.127		0.9778	0.119	
0.16	0.9742	0.105		0.9804	0.121		0.9757	0.116	
0.17	0.9720	0.103		0.9739	0.116		0.9738	0.113	
0.18	0.9706	0.100		0.9717	0.115		0.9724	0.110	
0.19	0.9699	0.097		0.9696	0.113		0.9717	0.105	
0.20	0.9677	0.093		0.9652	0.110		0.9706	0.101	
0.21	0.9656	0.091		0.9630	0.105		0.9684	0.099	
0.22	0.9656	0.090		0.9630	0.103		0.9658	0.097	
0.23	0.9648	0.088		0.9587	0.103		0.9648	0.094	
0.24	0.9641	0.085		0.9543	0.102		0.9633	0.093	
0.25	0.9605	0.084		0.9522	0.100		0.9615	0.091	
0.26	0.9605	0.081		0.9522	0.096		0.9604	0.089	
0.27	0.9605	0.079		0.9522	0.095		0.9597	0.086	Yes
0.28	0.9605	0.076		0.9500	0.093		0.9571	0.084	
0.29	0.9605	0.074	Yes	0.9500	0.091		0.9531	0.082	
0.30	0.9584	0.072		0.9478	0.090		0.9513	0.080	
0.31	0.9562	0.070		0.9478	0.088		0.9506	0.079	
0.32	0.9541	0.068		0.9478	0.087	Yes	0.9491	0.077	
0.33	0.9512	0.067		0.9457	0.086		0.9462	0.075	
0.34	0.9505	0.066		0.9435	0.085		0.9433	0.074	
0.35	0.9476	0.065		0.9413	0.084		0.9404	0.073	
0.36	0.9455	0.063		0.9413	0.081		0.9371	0.072	
0.37	0.9433	0.062		0.9370	0.077		0.9353	0.070	
0.38	0.9433	0.060		0.9348	0.074		0.9331	0.069	
0.39	0.9405	0.059		0.9326	0.073		0.9299	0.068	
0.40	0.9362	0.058		0.9283	0.073		0.9273	0.066	
0.41	0.9319	0.056		0.9283	0.072		0.9251	0.065	
0.42	0.9311	0.055		0.9239	0.071		0.9211	0.064	
0.43	0.9283	0.053		0.9217	0.070		0.9193	0.063	
0.44	0.9261	0.053		0.9196	0.069		0.9172	0.062	
0.45	0.9232	0.052		0.9196	0.067		0.9142	0.061	
0.46	0.9225	0.050		0.9174	0.066		0.9099	0.060	
0.47	0.9225	0.048		0.9174	0.066		0.9073	0.059	
0.48	0.9189	0.048		0.9152	0.064		0.9052	0.058	
0.49	0.9175	0.045		0.9152	0.063		0.9037	0.057	
0.50	0.9146	0.045		0.9152	0.062		0.9012	0.056	
0.51	0.9139	0.043		0.9109	0.062		0.8990	0.055	
0.52	0.9082	0.042		0.9109	0.061		0.8957	0.054	
0.53	0.9067	0.040		0.9109	0.060		0.8939	0.052	
0.54	0.9053	0.039		0.9065	0.060		0.8917	0.052	
0.55	0.9032	0.039		0.9022	0.060		0.8863	0.051	
0.56	0.9003	0.039		0.8978	0.060		0.8819	0.049	
0.57	0.8960	0.038		0.8957	0.059		0.8797	0.048	
0.58	0.8917	0.038		0.8870	0.059		0.8754	0.048	
0.59	0.8852	0.037		0.8826	0.058		0.8728	0.046	
0.60	0.8824	0.036		0.8783	0.057		0.8714	0.045	
0.61	0.8795	0.035		0.8761	0.055		0.8670	0.044	
0.62	0.8730	0.034		0.8761	0.055		0.8641	0.043	
0.63	0.8694	0.033		0.8696	0.051		0.8594	0.041	
0.64	0.8623	0.032		0.8652	0.051		0.8554	0.039	
0.65	0.8601	0.032		0.8652	0.051		0.8499	0.038	
0.66	0.8558	0.031		0.8609	0.048		0.8434	0.037	
0.67	0.8544	0.030		0.8565	0.047		0.8387	0.036	
0.68	0.8515	0.030		0.8543	0.047		0.8336	0.035	
0.69	0.8458	0.028		0.8500	0.045		0.8270	0.034	
0.70	0.8422	0.028		0.8457	0.045		0.8234	0.033	
0.71	0.8393	0.027		0.8391	0.044		0.8172	0.033	
0.72	0.8336	0.026		0.8348	0.039		0.8110	0.032	
0.73	0.8271	0.026		0.8261	0.037		0.8074	0.030	
0.74	0.8207	0.025		0.8217	0.037		0.7991	0.030	
0.75	0.8149	0.025		0.8152	0.037		0.7925	0.029	
0.76	0.8070	0.024		0.8043	0.036		0.7871	0.028	
0.77	0.7977	0.022		0.8000	0.035		0.7780	0.026	
0.78	0.7869	0.021		0.7891	0.034		0.7700	0.025	
0.79	0.7783	0.020		0.7761	0.033		0.7613	0.025	
0.80	0.7712	0.020		0.7609	0.033		0.7522	0.024	
0.81	0.7633	0.019		0.7522	0.033		0.7445	0.023	
0.82	0.7475	0.018		0.7478	0.031		0.7366	0.023	
0.83	0.7396	0.018		0.7413	0.031		0.7253	0.022	
0.84	0.7339	0.017		0.7261	0.030		0.7148	0.021	
0.85	0.7166	0.017		0.7109	0.028		0.7020	0.020	
0.86	0.7001	0.015		0.7043	0.028		0.6871	0.018	
0.87	0.6865	0.015		0.6870	0.027		0.6726	0.018	
0.88	0.6772	0.014		0.6717	0.024		0.6588	0.016	
0.89	0.6578	0.013		0.6630	0.023		0.6424	0.015	
0.90	0.6370	0.012		0.6370	0.020		0.6286	0.013	
0.91	0.6148	0.011		0.6196	0.019		0.6126	0.012	
0.92	0.5968	0.009		0.5870	0.019		0.5868	0.011	
0.93	0.5653	0.008		0.5717	0.016		0.5632	0.010	
0.94	0.5330	0.007		0.5348	0.015		0.5320	0.008	
0.95	0.4978	0.006		0.5022	0.013		0.4982	0.007	
0.96	0.4512	0.004		0.4609	0.012		0.4539	0.005	
0.97	0.3953	0.003		0.4022	0.008		0.3953	0.003	
0.98	0.3171	0.002		0.3304	0.005		0.3169	0.002	
0.99	0.2152	0.001		0.2152	0.004		0.2028	0.001	

Logistic regression - event_indicator 1 Cutoff Statistics

Cutoff	Training Accuracy	Training Sensitivity	Training Specificity	Training False Positive Rate	Training False Negative Rate	Kolmogorov-Smirnov (KS)	Training False Discovery Rate	Training F0.5 Score	Training F1 Score	Training Misclassification Rate (Event)	Validation Accuracy	Validation Sensitivity	Validation Specificity	Validation False Positive Rate	Validation False Negative Rate	Validation Kolmogorov-Smirnov (KS)	Validation False Discovery Rate	Validation F0.5 Score	Validation F1 Score	Validation Misclassification Rate (Event)	Test Accuracy	Test Sensitivity	Test Specificity	Test False Positive Rate	Test False Negative Rate	Test Kolmogorov-Smirnov (KS)	Test False Discovery Rate	Test F0.5 Score	Test F1 Score	Test Misclassification Rate (Event)
0.00	0.3027	1.0000	0.0000	1.0000	0.0000	0.0000	0.6973	0.3518	0.4647	0.6973	0.2997	1.0000	0.0000	1.0000	0.0000	0.0000	0.7003	0.3485	0.4612	0.7003	0.2988	1.0000	0.0000	1.0000	0.0000	0.0000	0.7012	0.3475	0.4601	0.7012
0.01	0.7946	1.0000	0.7054	0.2946	0.0000	0.7054	0.4043	0.6481	0.7467	0.2054	0.7844	1.0000	0.6921	0.3079	0.0000	0.6921	0.4185	0.6347	0.7354	0.2156	0.7825	0.9993	0.6902	0.3098	0.0007	0.6895	0.4212	0.6320	0.7330	0.2175
0.02	0.8304	0.9971	0.7580	0.2420	0.0029	0.7551	0.3586	0.6907	0.7807	0.1696	0.8163	0.9957	0.7395	0.2605	0.0043	0.7352	0.3794	0.6712	0.7646	0.1837	0.8214	0.9989	0.7458	0.2542	0.0011	0.7447	0.3739	0.6766	0.7697	0.1786
0.03	0.8482	0.9964	0.7839	0.2161	0.0036	0.7803	0.3332	0.7141	0.7990	0.1518	0.8313	0.9935	0.7619	0.2381	0.0065	0.7553	0.3590	0.6899	0.7792	0.1687	0.8419	0.9982	0.7754	0.2246	0.0018	0.7735	0.3456	0.7028	0.7905	0.1581
0.04	0.8610	0.9964	0.8022	0.1978	0.0036	0.7987	0.3137	0.7318	0.8128	0.1390	0.8456	0.9913	0.7833	0.2167	0.0087	0.7746	0.3382	0.7090	0.7937	0.1544	0.8558	0.9971	0.7956	0.2044	0.0029	0.7927	0.3248	0.7218	0.8052	0.1442
0.05	0.8721	0.9950	0.8187	0.1813	0.0050	0.8137	0.2956	0.7481	0.8249	0.1279	0.8573	0.9891	0.8009	0.1991	0.0109	0.7901	0.3199	0.7254	0.8060	0.1427	0.8639	0.9942	0.8083	0.1917	0.0058	0.8025	0.3115	0.7336	0.8136	0.1361
0.06	0.8810	0.9943	0.8318	0.1682	0.0057	0.8261	0.2804	0.7617	0.8349	0.1190	0.8632	0.9891	0.8093	0.1907	0.0109	0.7984	0.3106	0.7339	0.8125	0.1368	0.8730	0.9927	0.8220	0.1780	0.0073	0.8147	0.2962	0.7473	0.8236	0.1270
0.07	0.8879	0.9921	0.8427	0.1573	0.0079	0.8348	0.2675	0.7730	0.8428	0.1121	0.8704	0.9870	0.8205	0.1795	0.0130	0.8074	0.2983	0.7448	0.8202	0.1296	0.8800	0.9913	0.8326	0.1674	0.0087	0.8239	0.2838	0.7583	0.8316	0.1200
0.08	0.8925	0.9885	0.8508	0.1492	0.0115	0.8393	0.2579	0.7810	0.8477	0.1075	0.8743	0.9848	0.8270	0.1730	0.0152	0.8118	0.2911	0.7510	0.8244	0.1257	0.8864	0.9898	0.8424	0.1576	0.0102	0.8322	0.2720	0.7686	0.8389	0.1136
0.09	0.8971	0.9871	0.8580	0.1420	0.0129	0.8451	0.2489	0.7888	0.8531	0.1029	0.8801	0.9848	0.8353	0.1647	0.0152	0.8201	0.2810	0.7601	0.8312	0.1199	0.8902	0.9880	0.8486	0.1514	0.0120	0.8366	0.2645	0.7751	0.8432	0.1098
0.10	0.8999	0.9842	0.8633	0.1367	0.0158	0.8475	0.2424	0.7942	0.8562	0.1001	0.8866	0.9826	0.8456	0.1544	0.0174	0.8282	0.2686	0.7708	0.8386	0.1134	0.8953	0.9858	0.8568	0.1432	0.0142	0.8426	0.2543	0.7839	0.8491	0.1047
0.11	0.9047	0.9806	0.8717	0.1283	0.0194	0.8523	0.2316	0.8032	0.8616	0.0953	0.8906	0.9826	0.8512	0.1488	0.0174	0.8338	0.2614	0.7772	0.8433	0.1094	0.8987	0.9844	0.8622	0.1378	0.0156	0.8466	0.2473	0.7899	0.8531	0.1013
0.12	0.9079	0.9806	0.8764	0.1236	0.0194	0.8570	0.2251	0.8089	0.8657	0.0921	0.8938	0.9826	0.8558	0.1442	0.0174	0.8384	0.2554	0.7825	0.8472	0.1062	0.9010	0.9826	0.8662	0.1338	0.0174	0.8488	0.2422	0.7942	0.8557	0.0990
0.13	0.9105	0.9785	0.8810	0.1190	0.0215	0.8595	0.2188	0.8140	0.8688	0.0895	0.8990	0.9826	0.8633	0.1367	0.0174	0.8459	0.2454	0.7913	0.8536	0.1010	0.9044	0.9804	0.8720	0.1280	0.0196	0.8523	0.2346	0.8005	0.8596	0.0956
0.14	0.9131	0.9770	0.8854	0.1146	0.0230	0.8624	0.2127	0.8191	0.8720	0.0869	0.9010	0.9826	0.8660	0.1340	0.0174	0.8487	0.2416	0.7947	0.8561	0.0990	0.9075	0.9793	0.8769	0.1231	0.0207	0.8562	0.2278	0.8063	0.8635	0.0925
0.15	0.9155	0.9756	0.8894	0.1106	0.0244	0.8651	0.2070	0.8238	0.8749	0.0845	0.9049	0.9804	0.8726	0.1274	0.0196	0.8530	0.2330	0.8019	0.8607	0.0951	0.9100	0.9778	0.8811	0.1189	0.0222	0.8589	0.2220	0.8111	0.8665	0.0900
0.16	0.9190	0.9742	0.8950	0.1050	0.0258	0.8692	0.1988	0.8307	0.8792	0.0810	0.9094	0.9804	0.8791	0.1209	0.0196	0.8595	0.2238	0.8100	0.8665	0.0906	0.9112	0.9757	0.8837	0.1163	0.0243	0.8594	0.2186	0.8138	0.8678	0.0888
0.17	0.9197	0.9720	0.8969	0.1031	0.0280	0.8689	0.1963	0.8325	0.8799	0.0803	0.9107	0.9739	0.8837	0.1163	0.0261	0.8576	0.2182	0.8140	0.8674	0.0893	0.9130	0.9738	0.8871	0.1129	0.0262	0.8610	0.2138	0.8177	0.8700	0.0870
0.18	0.9212	0.9706	0.8997	0.1003	0.0294	0.8703	0.1922	0.8358	0.8817	0.0788	0.9107	0.9717	0.8847	0.1153	0.0283	0.8564	0.2172	0.8145	0.8671	0.0893	0.9148	0.9724	0.8902	0.1098	0.0276	0.8626	0.2095	0.8213	0.8721	0.0852
0.19	0.9233	0.9699	0.9031	0.0969	0.0301	0.8730	0.1870	0.8402	0.8845	0.0767	0.9121	0.9696	0.8874	0.1126	0.0304	0.8570	0.2134	0.8174	0.8685	0.0879	0.9177	0.9717	0.8947	0.1053	0.0283	0.8664	0.2027	0.8269	0.8759	0.0823
0.20	0.9253	0.9677	0.9069	0.0931	0.0323	0.8746	0.1814	0.8446	0.8869	0.0747	0.9127	0.9652	0.8902	0.1098	0.0348	0.8554	0.2100	0.8198	0.8689	0.0873	0.9201	0.9706	0.8986	0.1014	0.0294	0.8692	0.1969	0.8318	0.8789	0.0799
0.21	0.9260	0.9656	0.9088	0.0912	0.0344	0.8743	0.1788	0.8465	0.8876	0.0740	0.9153	0.9630	0.8949	0.1051	0.0370	0.8579	0.2032	0.8253	0.8720	0.0847	0.9212	0.9684	0.9011	0.0989	0.0316	0.8695	0.1934	0.8345	0.8801	0.0788
0.22	0.9270	0.9656	0.9103	0.0897	0.0344	0.8759	0.1763	0.8487	0.8890	0.0730	0.9166	0.9630	0.8967	0.1033	0.0370	0.8598	0.2004	0.8277	0.8738	0.0834	0.9220	0.9658	0.9034	0.0966	0.0342	0.8692	0.1901	0.8369	0.8810	0.0780
0.23	0.9283	0.9648	0.9125	0.0875	0.0352	0.8773	0.1728	0.8515	0.8907	0.0717	0.9153	0.9587	0.8967	0.1033	0.0413	0.8554	0.2011	0.8265	0.8715	0.0847	0.9234	0.9648	0.9057	0.0943	0.0352	0.8705	0.1866	0.8398	0.8826	0.0766
0.24	0.9301	0.9641	0.9153	0.0847	0.0359	0.8794	0.1683	0.8552	0.8930	0.0699	0.9147	0.9543	0.8977	0.1023	0.0457	0.8520	0.2004	0.8264	0.8702	0.0853	0.9241	0.9633	0.9074	0.0926	0.0367	0.8707	0.1841	0.8417	0.8835	0.0759
0.25	0.9296	0.9605	0.9162	0.0838	0.0395	0.8768	0.1673	0.8555	0.8921	0.0704	0.9160	0.9522	0.9005	0.0995	0.0478	0.8526	0.1963	0.8295	0.8716	0.0840	0.9249	0.9615	0.9093	0.0907	0.0385	0.8708	0.1813	0.8438	0.8844	0.0751
0.26	0.9316	0.9605	0.9190	0.0810	0.0395	0.8796	0.1626	0.8594	0.8948	0.0684	0.9186	0.9522	0.9042	0.0958	0.0478	0.8564	0.1904	0.8346	0.8751	0.0814	0.9260	0.9604	0.9113	0.0887	0.0396	0.8717	0.1782	0.8462	0.8857	0.0740
0.27	0.9327	0.9605	0.9206	0.0794	0.0395	0.8811	0.1600	0.8616	0.8963	0.0673	0.9192	0.9522	0.9051	0.0949	0.0478	0.8573	0.1889	0.8359	0.8760	0.0808	0.9276	0.9597	0.9139	0.0861	0.0403	0.8736	0.1739	0.8497	0.8879	0.0724
0.28	0.9349	0.9605	0.9237	0.0763	0.0395	0.8842	0.1547	0.8661	0.8993	0.0651	0.9199	0.9500	0.9070	0.0930	0.0500	0.8570	0.1862	0.8378	0.8766	0.0801	0.9280	0.9571	0.9156	0.0844	0.0429	0.8727	0.1714	0.8514	0.8882	0.0720
0.29	0.9366	0.9605	0.9262	0.0738	0.0395	0.8867	0.1504	0.8697	0.9017	0.0634	0.9212	0.9500	0.9088	0.0912	0.0500	0.8588	0.1832	0.8404	0.8784	0.0788	0.9282	0.9531	0.9176	0.0824	0.0469	0.8708	0.1686	0.8532	0.8881	0.0718
0.30	0.9372	0.9584	0.9281	0.0719	0.0416	0.8865	0.1474	0.8718	0.9024	0.0628	0.9212	0.9478	0.9098	0.0902	0.0522	0.8576	0.1820	0.8410	0.8781	0.0788	0.9293	0.9513	0.9200	0.0800	0.0487	0.8713	0.1649	0.8560	0.8894	0.0707
0.31	0.9377	0.9562	0.9296	0.0704	0.0438	0.8859	0.1450	0.8735	0.9028	0.0623	0.9225	0.9478	0.9116	0.0884	0.0522	0.8595	0.1789	0.8437	0.8799	0.0775	0.9299	0.9506	0.9210	0.0790	0.0494	0.8716	0.1631	0.8574	0.8901	0.0701
0.32	0.9385	0.9541	0.9318	0.0682	0.0459	0.8859	0.1414	0.8762	0.9038	0.0615	0.9238	0.9478	0.9135	0.0865	0.0522	0.8613	0.1758	0.8463	0.8817	0.0762	0.9307	0.9491	0.9229	0.0771	0.0509	0.8720	0.1601	0.8597	0.8912	0.0693
0.33	0.9388	0.9512	0.9334	0.0666	0.0488	0.8846	0.1390	0.8777	0.9039	0.0612	0.9238	0.9457	0.9144	0.0856	0.0543	0.8601	0.1746	0.8470	0.8815	0.0762	0.9315	0.9462	0.9252	0.0748	0.0538	0.8714	0.1565	0.8623	0.8919	0.0685
0.34	0.9392	0.9505	0.9343	0.0657	0.0495	0.8848	0.1374	0.8789	0.9044	0.0608	0.9238	0.9435	0.9153	0.0847	0.0565	0.8588	0.1733	0.8477	0.8812	0.0762	0.9314	0.9433	0.9263	0.0737	0.0567	0.8696	0.1549	0.8630	0.8915	0.0686
0.35	0.9388	0.9476	0.9349	0.0651	0.0524	0.8825	0.1366	0.8790	0.9036	0.0612	0.9238	0.9413	0.9163	0.0837	0.0587	0.8576	0.1721	0.8484	0.8810	0.0762	0.9313	0.9404	0.9274	0.0726	0.0596	0.8678	0.1534	0.8638	0.8910	0.0687
0.36	0.9394	0.9455	0.9368	0.0632	0.0545	0.8823	0.1335	0.8813	0.9043	0.0606	0.9257	0.9413	0.9191	0.0809	0.0587	0.8604	0.1673	0.8524	0.8837	0.0743	0.9308	0.9371	0.9282	0.0718	0.0629	0.8653	0.1525	0.8640	0.8901	0.0692
0.37	0.9394	0.9433	0.9377	0.0623	0.0567	0.8810	0.1320	0.8821	0.9041	0.0606	0.9270	0.9370	0.9228	0.0772	0.0630	0.8597	0.1615	0.8565	0.8850	0.0730	0.9313	0.9353	0.9296	0.0704	0.0647	0.8649	0.1502	0.8656	0.8905	0.0687
0.38	0.9409	0.9433	0.9399	0.0601	0.0567	0.8832	0.1280	0.8854	0.9063	0.0591	0.9283	0.9348	0.9256	0.0744	0.0652	0.8604	0.1569	0.8600	0.8866	0.0717	0.9317	0.9331	0.9311	0.0689	0.0669	0.8642	0.1477	0.8673	0.8909	0.0683
0.39	0.9412	0.9405	0.9415	0.0585	0.0595	0.8819	0.1254	0.8870	0.9063	0.0588	0.9283	0.9326	0.9265	0.0735	0.0674	0.8591	0.1555	0.8608	0.8864	0.0717	0.9317	0.9299	0.9325	0.0675	0.0701	0.8624	0.1456	0.8685	0.8906	0.0683
0.40	0.9405	0.9362	0.9424	0.0576	0.0638	0.8785	0.1242	0.8873	0.9050	0.0595	0.9277	0.9283	0.9274	0.0726	0.0717	0.8557	0.1545	0.8609	0.8850	0.0723	0.9318	0.9273	0.9337	0.0663	0.0727	0.8611	0.1436	0.8697	0.8904	0.0682
0.41	0.9403	0.9319	0.9439	0.0561	0.0681	0.8758	0.1217	0.8885	0.9043	0.0597	0.9283	0.9283	0.9284	0.0716	0.0717	0.8566	0.1528	0.8623	0.8859	0.0717	0.9318	0.9251	0.9347	0.0653	0.0749	0.8598	0.1422	0.8705	0.8902	0.0682
0.42	0.9412	0.9311	0.9455	0.0545	0.0689	0.8766	0.1188	0.8907	0.9055	0.0588	0.9277	0.9239	0.9293	0.0707	0.0761	0.8532	0.1517	0.8624	0.8845	0.0723	0.9315	0.9211	0.9359	0.0641	0.0789	0.8571	0.1404	0.8713	0.8893	0.0685
0.43	0.9412	0.9283	0.9467	0.0533	0.0717	0.8750	0.1167	0.8919	0.9052	0.0588	0.9277	0.9217	0.9302	0.0698	0.0783	0.8520	0.1503	0.8632	0.8843	0.0723	0.9317	0.9193	0.9370	0.0630	0.0807	0.8563	0.1386	0.8724	0.8894	0.0683
0.44	0.9409	0.9261	0.9474	0.0526	0.0739	0.8735	0.1158	0.8923	0.9047	0.0591	0.9277	0.9196	0.9312	0.0688	0.0804	0.8507	0.1489	0.8640	0.8840	0.0723	0.9318	0.9172	0.9381	0.0619	0.0828	0.8552	0.1368	0.8735	0.8894	0.0682
0.45	0.9405	0.9232	0.9480	0.0520	0.0768	0.8712	0.1149	0.8925	0.9038	0.0595	0.9290	0.9196	0.9330	0.0670	0.0804	0.8526	0.1455	0.8668	0.8859	0.0710	0.9317	0.9142	0.9392	0.0608	0.0858	0.8534	0.1351	0.8743	0.8889	0.0683
0.46	0.9420	0.9225	0.9505	0.0495	0.0775	0.8730	0.1100	0.8963	0.9060	0.0580	0.9290	0.9174	0.9340	0.0660	0.0826	0.8513	0.1440	0.8676	0.8856	0.0710	0.9307	0.9099	0.9396	0.0604	0.0901	0.8495	0.1348	0.8738	0.8870	0.0693
0.47	0.9429	0.9225	0.9517	0.0483	0.0775	0.8743	0.1076	0.8983	0.9072	0.0571	0.9290	0.9174	0.9340	0.0660	0.0826	0.8513	0.1440	0.8676	0.8856	0.0710	0.9311	0.9073	0.9412	0.0588	0.0927	0.8485	0.1321	0.8755	0.8872	0.0689
0.48	0.9422	0.9189	0.9524	0.0476	0.0811	0.8713	0.1067	0.8983	0.9059	0.0578	0.9296	0.9152	0.9358	0.0642	0.0848	0.8510	0.1408	0.8698	0.8863	0.0704	0.9307	0.9052	0.9416	0.0584	0.0948	0.8468	0.1315	0.8756	0.8865	0.0693
0.49	0.9433	0.9175	0.9545	0.0455	0.0825	0.8720	0.1025	0.9015	0.9074	0.0567	0.9303	0.9152	0.9367	0.0633	0.0848	0.8520	0.1391	0.8713	0.8872	0.0697	0.9312	0.9037	0.9429	0.0571	0.0963	0.8466	0.1292	0.8772	0.8869	0.0688
0.50	0.9429	0.9146	0.9552	0.0448	0.0854	0.8698	0.1015	0.9017	0.9065	0.0571	0.9309	0.9152	0.9377	0.0623	0.0848	0.8529	0.1373	0.8727	0.8882	0.0691	0.9314	0.9012	0.9443	0.0557	0.0988	0.8454	0.1268	0.8787	0.8870	0.0686
0.51	0.9438	0.9139	0.9567	0.0433	0.0861	0.8706	0.0984	0.9041	0.9077	0.0562	0.9296	0.9109	0.9377	0.0623	0.0891	0.8485	0.1379	0.8715	0.8858	0.0704	0.9313	0.8990	0.9450	0.0550	0.1010	0.8440	0.1255	0.8793	0.8866	0.0687
0.52	0.9429	0.9082	0.9580	0.0420	0.0918	0.8661	0.0964	0.9045	0.9059	0.0571	0.9303	0.9109	0.9386	0.0614	0.0891	0.8495	0.1361	0.8729	0.8868	0.0697	0.9313	0.8957	0.9464	0.0536	0.1043	0.8421	0.1231	0.8806	0.8862	0.0687
0.53	0.9435	0.9067	0.9595	0.0405	0.0933	0.8663	0.0933	0.9067	0.9067	0.0565	0.9309	0.9109	0.9395	0.0605	0.0891	0.8504	0.1343	0.8744	0.8877	0.0691	0.9315	0.8939	0.9475	0.0525	0.1061	0.8414	0.1211	0.8818	0.8863	0.0685
0.54	0.9440	0.9053	0.9608	0.0392	0.0947	0.8661	0.0908	0.9084	0.9073	0.0560	0.9296	0.9065	0.9395	0.0605	0.0935	0.8461	0.1349	0.8731	0.8854	0.0704	0.9313	0.8917	0.9481	0.0519	0.1083	0.8398	0.1201	0.8822	0.8858	0.0687
0.55	0.9435	0.9032	0.9611	0.0389	0.0968	0.8642	0.0903	0.9084	0.9064	0.0565	0.9290	0.9022	0.9405	0.0595	0.0978	0.8426	0.1336	0.8733	0.8839	0.0710	0.9304	0.8863	0.9492	0.0508	0.1137	0.8355	0.1185	0.8824	0.8839	0.0696
0.56	0.9429	0.9003	0.9614	0.0386	0.0997	0.8617	0.0899	0.9081	0.9052	0.0571	0.9277	0.8978	0.9405	0.0595	0.1022	0.8383	0.1342	0.8720	0.8815	0.0723	0.9301	0.8819	0.9506	0.0494	0.1181	0.8325	0.1162	0.8834	0.8829	0.0699
0.57	0.9418	0.8960	0.9617	0.0383	0.1040	0.8577	0.0897	0.9074	0.9031	0.0582	0.9277	0.8957	0.9414	0.0586	0.1043	0.8370	0.1326	0.8729	0.8813	0.0723	0.9302	0.8797	0.9517	0.0483	0.1203	0.8314	0.1142	0.8846	0.8828	0.0698
0.58	0.9409	0.8917	0.9623	0.0377	0.1083	0.8540	0.0887	0.9073	0.9014	0.0591	0.9251	0.8870	0.9414	0.0586	0.1130	0.8284	0.1338	0.8703	0.8765	0.0749	0.9294	0.8754	0.9525	0.0475	0.1246	0.8278	0.1130	0.8846	0.8811	0.0706
0.59	0.9392	0.8852	0.9626	0.0374	0.1148	0.8479	0.0886	0.9060	0.8981	0.0608	0.9244	0.8826	0.9423	0.0577	0.1174	0.8249	0.1325	0.8705	0.8750	0.0756	0.9295	0.8728	0.9537	0.0463	0.1272	0.8265	0.1107	0.8860	0.8810	0.0705
0.60	0.9392	0.8824	0.9639	0.0361	0.1176	0.8462	0.0862	0.9073	0.8978	0.0608	0.9238	0.8783	0.9433	0.0567	0.1217	0.8215	0.1312	0.8707	0.8735	0.0762	0.9302	0.8714	0.9553	0.0447	0.1286	0.8266	0.1076	0.8881	0.8818	0.0698
0.61	0.9390	0.8795	0.9648	0.0352	0.1205	0.8443	0.0844	0.9081	0.8972	0.0610	0.9244	0.8761	0.9451	0.0549	0.1239	0.8212	0.1277	0.8731	0.8742	0.0756	0.9293	0.8670	0.9559	0.0441	0.1330	0.8229	0.1067	0.8879	0.8800	0.0707
0.62	0.9379	0.8730	0.9661	0.0339	0.1270	0.8391	0.0822	0.9085	0.8949	0.0621	0.9244	0.8761	0.9451	0.0549	0.1239	0.8212	0.1277	0.8731	0.8742	0.0756	0.9291	0.8641	0.9568	0.0432	0.1359	0.8209	0.1050	0.8886	0.8793	0.0709
0.63	0.9372	0.8694	0.9667	0.0333	0.1306	0.8361	0.0811	0.9085	0.8935	0.0628	0.9251	0.8696	0.9488	0.0512	0.1304	0.8184	0.1209	0.8772	0.8743	0.0749	0.9292	0.8594	0.9590	0.0410	0.1406	0.8183	0.1008	0.8910	0.8789	0.0708
0.64	0.9357	0.8623	0.9676	0.0324	0.1377	0.8299	0.0796	0.9081	0.8904	0.0643	0.9238	0.8652	0.9488	0.0512	0.1348	0.8141	0.1214	0.8759	0.8719	0.0762	0.9292	0.8554	0.9607	0.0393	0.1446	0.8161	0.0974	0.8927	0.8784	0.0708
0.65	0.9355	0.8601	0.9682	0.0318	0.1399	0.8283	0.0784	0.9086	0.8898	0.0645	0.9238	0.8652	0.9488	0.0512	0.1348	0.8141	0.1214	0.8759	0.8719	0.0762	0.9285	0.8499	0.9619	0.0381	0.1501	0.8118	0.0952	0.8933	0.8765	0.0715
0.66	0.9344	0.8558	0.9685	0.0315	0.1442	0.8244	0.0781	0.9079	0.8876	0.0656	0.9244	0.8609	0.9516	0.0484	0.1391	0.8125	0.1161	0.8792	0.8722	0.0756	0.9274	0.8434	0.9632	0.0368	0.1566	0.8065	0.0930	0.8935	0.8740	0.0726
0.67	0.9351	0.8544	0.9701	0.0299	0.1456	0.8245	0.0746	0.9103	0.8885	0.0649	0.9238	0.8565	0.9526	0.0474	0.1435	0.8091	0.1146	0.8795	0.8707	0.0762	0.9267	0.8387	0.9642	0.0358	0.1613	0.8029	0.0910	0.8940	0.8724	0.0733
0.68	0.9344	0.8515	0.9704	0.0296	0.1485	0.8219	0.0741	0.9100	0.8871	0.0656	0.9231	0.8543	0.9526	0.0474	0.1457	0.8069	0.1149	0.8788	0.8695	0.0769	0.9257	0.8336	0.9650	0.0350	0.1664	0.7986	0.0897	0.8939	0.8703	0.0743
0.69	0.9338	0.8458	0.9720	0.0280	0.1542	0.8177	0.0709	0.9111	0.8855	0.0662	0.9238	0.8500	0.9553	0.0447	0.1500	0.8053	0.1093	0.8822	0.8699	0.0762	0.9248	0.8270	0.9664	0.0336	0.1730	0.7934	0.0870	0.8944	0.8679	0.0752
0.70	0.9329	0.8422	0.9723	0.0277	0.1578	0.8145	0.0705	0.9106	0.8837	0.0671	0.9225	0.8457	0.9553	0.0447	0.1543	0.8010	0.1098	0.8809	0.8673	0.0775	0.9239	0.8234	0.9667	0.0333	0.1766	0.7901	0.0867	0.8938	0.8660	0.0761
0.71	0.9322	0.8393	0.9726	0.0274	0.1607	0.8119	0.0700	0.9104	0.8824	0.0678	0.9212	0.8391	0

Logistic regression - event_indicator 1 Misclassification

Response	Event	Value	Training Observations	Validation Observations	Test Observations
Correct	1	True Positive	1,275	421	2,480
Incorrect	1	False Negative	119	39	272
Correct	0	True Negative	3,067	1,008	6,099
Incorrect	0	False Positive	144	67	360

Logistic regression - event_indicator 1 Assessment Statistics

Partition	KS (Youden)	Misclassification Rate	Misclassification Rate (Event)	C Statistic	False Positive Rate	False Discovery Rate	F1 Score	Lift	Cumulative Lift	Cumulative % Events	Cumulative % Captured	Gain	Gini	Gamma	Tau	Observations Used	Unused
Training	0.8867	0.0571	0.0571	0.984	0.045	0.101	0.907	3.300	3.300	99.567	16.499	2.300	0.967	0.969	0.408	4,605	0
Validation	0.8613	0.0691	0.0691	0.975	0.062	0.137	0.888	3.174	3.174	94.805	15.870	2.174	0.950	0.952	0.399	1,535	0
Test	0.8736	0.0686	0.0686	0.980	0.056	0.127	0.887	3.328	3.328	99.349	16.642	2.328	0.960	0.962	0.402	9,211	0

Logistic regression - event_indicator 1 Assessment Summary

Plot	Summary
Confusion Matrix	The confusion matrix plot displays the number of observations predicting each response level. A greater number of observations where the observed level and predicted level are the same indicates a better model. For this data, the percentages of each observed value that are correctly predicted in the validation partition are as follows: 0 - 93.77% and 1 - 91.52%.
Lift	The lift plot measures the ratio of percent captured response to the baseline percent response. The validation partition has a lift of 3.1739 at the 5% quantile meaning there are about 3.17 times more events in that quantile than expected by random (5% of the total number of events).
Cumulative Lift	Cumulative lift measures the ratio of percent captured response to the baseline percent response, up to and including the specified quantile. The validation partition has a cumulative lift of 3.2391 in the 10% quantile meaning there are about 3.24 times more events in the first two quantiles than expected by random (10% of the total number of events). Because this value is greater than 1, it is better to use your model to identify responders than no model, based on the validation partition.
ROC	The receiver operator characteristic (ROC) is a plot of sensitivity (the true positive rate) against 1-specificity (the false positive rate), which are both measures of classification based on the confusion matrix. These measures are calculated at various cutoff values. To help identify the best cutoff to use when scoring your data, the KS cutoff reference line is drawn at the value of 1-specificity where the greatest difference between sensitivity and 1-specificity is observed for the validation partition. The KS cutoff line is drawn at the cutoff value 0.32 where the 1-specificity value is 0.087 and the sensitivity value is 0.948. Cutoff values range from 0 to 0.99, inclusive, in increments of 0.01. At each cutoff value, the predicted response classification is determined by whether the predicted probability of the response event_indicator being 1 is greater than or equal to the cutoff value. When the predicted probability of the event is greater than or equal to the cutoff value, then the predicted classification is 1, otherwise it is NOT 1.
Cutoff	The cutoff plot shows how different model assessment statistics change as the prediction cutoff value changes. The model assessment statistics are based on the selected event for the model compared to non-events. You can interactively move the cutoff line to represent different prediction cutoff values. As you move the cutoff line, the model assessment statistics are updated. This allows you to choose a cutoff that best represents your particular problem and business objective. When the accuracy statistic is enabled, its value for the selected cutoff is always displayed.
Misclassification	The misclassification plot is a visual representation of the accuracy of the prediction at the specified cutoff value, 0.50. The plot displays the number of true positives for events that are correctly classified, false positives for NOT events that are classified as events, false negatives for events that are classified as NOT events, and true negatives for NOT events that are classified as NOT events. True negatives include NOT event classifications that predict a different level from observed, as long as both are NOT events. The predicted response classification is determined by whether the predicted probability of the level 1 for the response event_indicator is greater than or equal to the cutoff value. When it is greater than or equal to the cutoff value, the predicted classification is an event, otherwise it is a NOT event. For this data, for the bar corresponding to the event level of event_indicator, 1, the segment of the bar colored as "Correct" corresponds to true positives.

Cluster - id_current_fs_relationship 1 Centroids

Cluster ID	id_current_fs_relationship	id_direct_contact	id_important_activity
1	true	false	false
2	false	false	false

Cluster - id_current_fs_relationship 1 Cluster Summary

Cluster ID	Observations	Min centroid-to-observati...	Max centroid-to-observati...	Nearest Cluster	Centroid Distance	Average Distance	Within Distance
1	5,805	0	2	2	1	0.0999138674	580
2	9,546	0	2	1	1	0.0977372722	933

Cluster - id_current_fs_relationship 1 Model Information

Description	Value
Clustering Algorithm	K-modes
Maximum Iterations	150
Stop Criterion	Cluster Change
Stop Criterion Value	0
Clusters	5
Initialization	Forgy
Seed	1,234
Distance for Nominal Variables	Binary
Nominal Imputation	None

Cluster - id_current_fs_relationship 1 Iteration History

Iteration	Sum of Distance	Distance Change	Stop Criterion
0	1513	.	.
1	1513	0	0

Cluster - id_current_fs_relationship 1 Frequency

Variable	Level	Frequency	1	2
id_current_fs_relationship	false	9546	0	9,546
id_current_fs_relationship	true	5805	5,805	0
id_direct_contact	false	14614	5,513	9,101
id_direct_contact	true	737	292	445
id_important_activity	false	14575	5,517	9,058
id_important_activity	true	776	288	488

Cluster - id_current_fs_relationship 1 Parallel Coordinates Plot

Cluster ID	id_current_fs_relationship	id_direct_contact	id_important_activity	Frequency
2	false	false	false	8,641
1	true	true	true	9
1	true	false	true	279
1	true	true	false	283
2	false	false	true	460
1	true	false	false	5,234
2	false	true	true	28
2	false	true	false	417

Automated prediction - event_indicator 1 Prediction Description

1. Select response for Automated Prediction.	A report author selected event_indicator as the response.
2. Find the champion model.	The prediction uses a Gradient Boosting champion model. The model was automatically selected from a comparison of Decision Tree, Linear Regression, and Gradient Boosting models. The model was chosen based on the lowest Average Squared Error (0.0336). Average Squared Error measures how different the model's estimates are from actual values in the data.
3. Predict outcome.	Given the user's inputs, predict the response using the champion model.

Automated prediction - event_indicator 1 Relative Importance

Variable	Importance
net_worth	1.0000
credit_score	0.7126
num_dependents	0.6839
at_current_job_1_year	0.6791
credit_history_mos	0.4058
job_in_education	0.3867
num_transactions	0.2306
debt_to_income	0.1961
amount	0.1806
gender	0.1296
age	0.0888
job_in_hospitality	0.0780
job_in_financial	0.0477
marital_status_single	0.0224
job_in_media	0.0065